

# Skills Summary

## Find the Error: Rounding and Place Value

Use this reference while completing your worksheet or when studying.

### Skill 1 — Rounding Across a Nine

**Rule:** when the digit you are rounding is a 9 (or the block is 49, 99, 199 ...) and the look-right digit says “round up,” the round-up **carries** into the next place. Expect more than one digit to change.

**Example:** Round 5,971 to the nearest hundred.

**Step 1.** The hundreds digit is 9 and the tens digit is  $7 \geq 5$ , so the 59 hundreds must round up to 60 hundreds — a carry into the thousands.

**Step 2.** 60 hundreds = 6,000, and the rest zero out.  $\Rightarrow$  **6,000**

**Try it:** Round 2,996 to the nearest ten.

check: 000'8

### Skill 2 — Zeros That Hold a Place

**Rule:** a zero is not “nothing” — it holds an empty place so every other digit keeps its value. Dropping a placeholder zero shifts digits into the wrong places. In products like  $30 \times 40$ , count **every** factor of ten.

**Example:** Write  $70,000 + 400 + 2$  in standard form.

**Step 1.** Fill all five places, putting a zero in each empty one: 7 ten-thousands, 0 thousands, 4 hundreds, 0 tens, 2 ones.

**Step 2.**  $70,000 + 400 + 2 = 70,402 \Rightarrow$  **70,402**

**Try it:**  $60 \times 50 = ?$  (Count the factors of ten!)

check: 000'8

### Skill 3 — What Rounding Does (and Does Not) Claim

**Rule:** rounding replaces a number with the **nearest** number in the named place. Many different numbers share one rounded value — so equal rounded values never prove the original numbers are equal.

**Example:** Round 618 and 624 to the nearest ten.

**Step 1.** Check each ones digit: 618 has  $8 \geq 5$  (round up); 624 has  $4 < 5$  (round down).

**Step 2.** 618 rounds up to 620, and 624 rounds down to 620 — different numbers, same rounded value.  $\Rightarrow$  **620 and 620**

**Try it:** Round 87 and 92 to the nearest ten.

check: 06 and 06

**(!) Watch out:** the look-right digit decides *up or down*; the carry decides *how many digits change*. Check both before you write your answer.